

Efficiency Evaluation of Scalable Multiply and Accumulate Architectures in DSP: A Comparative Study of LUT Based and LUT-Less Based Approaches

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Abstract— Multiply and Accumulate (MAC) units are widely used in digital signal processing. They are crucial for activities like correlation, convolution, and filtering, which allow for fast and effective signal processing. A fundamental component in modern digital signal processing systems, these scalable units serve as the foundation for a wide variety of DSP algorithms. The distributed arithmetic method, which offers reduced computational complexity while preserving accuracy through parallelism and distributed multiplication operations, is a highly effective way for creating Multiply and Accumulate (MAC) structures in digital signal processing. A scalable Multiply and Accumulate (MAC) architectures are shown in this article. Also, explored the design of multiply and accumulate structures at various bit-widths: 4, 8, 16, 32 and 64-bit versions, as well as analyzed LUT-based and LUT-less approaches. Significant variations in space and power consumption between LUT-based and LUT-less systems are highlighted by the data in the article for a variety of bit-width configurations. A quantitative examination shows that whereas LUT-less devices continue to have a smaller footprint and lower power consumption even on larger bit-widths, LUT-based solutions clearly consume large area and consumes more power as width of the bits expands. The proposed method is designed and simulated using Xilinx Vivado 2022.2 software.

Keywords- Multiply and Accumulate (MAC), Distributed Arithmetic (DA), LUT, LUT-Less, Xilinx Vivado 2022.2.

I. INTRODUCTION

For a variety of real-time applications, multiply and accumulate [1] (MAC) units—essential parts of specialized processors and digital signal processors (DSPs)—are needed as shown in Fig.1. The MAC device multiplies two numbers quickly and adds the result to an accumulator, showcasing exceptional speed and efficiency. These applications cover a wide range of industries, such as scientific computing, control systems, medical imaging, wireless communication, radar and sonar systems, audio and image processing, and video compression. The main reasons MAC units are so popular in these applications are because of their precision, speed, and adaptability for real-time or almost real-time processing. A level of precision provided by MAC units is essential for operations

like scientific simulations and medical imaging. Furthermore, they are perfect for applications in portable and energy-constrained devices because to their low power consumption and parallelism, which guarantees their continued importance in signal processing and computing tasks across multiple domains.

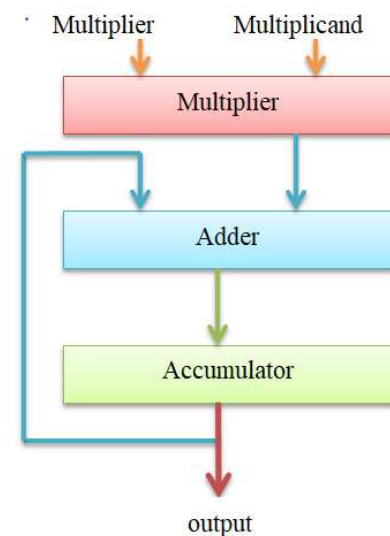


Fig. 1. MAC Architecture

In signal processing, image processing, and other applications, correlation and convolution-related functions are critical, and multiply and accumulate (MAC) operations play a key role in these domains. Sliding a kernel or filter over the input data, multiplying the associated elements, and adding the results are the steps involved in correlation and convolution procedures. These activities revolve around MAC units, which effectively manage the phases of multiplication and accumulation. While MAC units apply filters to input data in convolution [2] to enable tasks like image filtering and feature extraction, they also compute the similarity between signals in tasks like cross-correlation and auto-correlation. In order to handle input

efficiently, MAC operations are also used by convolutional layers in convolutional neural networks (CNNs) and by finite impulse response (FIR) filters. Because of their speed, accuracy, and parallelism, MAC units are highly useful for high-performance, real-time processing [3] in activities linked to correlation and convolution, which guarantees their widespread application across a variety of industries.

II. LITERATURE SURVEY

The design and application of a specialized DSP core made for MP3 and AAC audio codecs is the primary goal of Kyoung Ho Bang et al. This paper provides insights into effective audio processing solutions for consumer electronics applications, making it especially useful for researchers and engineers working in the fields of audio processing and VLSI design.[4].

The novel FACCUC approach is introduced in the publication "FACCUC: Enable Fast Accumulation for High-Speed DSP Systems," which was written by M. Wang et al. Its goal is to improve the effectiveness of accumulation operations in high-speed DSP systems. Because of its recent release, this work provides scholars looking for current information on high-speed signal processing and DSP system optimization with insightful information. It offers a potentially significant addition to the computational efficiency in DSP systems.[5]

"Near-Precise Parameter Approximation for Multiple Multiplications on a Single DSP Block," by E. Kalali et al. The study presents a novel approach to provide near-precise parameter estimate for numerous multiplications on a single DSP block. This work adds much to the fields of resource-efficient computation and DSP optimisation, offering academics interested in the most recent developments in the area up-to-date insights.[6]

TiNNA[7] is a small accelerator with optimized DSP for better efficiency that is intended for neural network processing. More information about this technique may be found in the publication written by X. Hu et al. Noteworthy contribution to the fields of hardware optimization and neural network accelerators, providing information on small and effective hardware solutions for AI processing. An invaluable resource for scholars curious about the most recent advancements in DSP optimization and AI hardware acceleration.

W. Li and colleagues' study, "Low-Complexity Precision-Scalable Multiply-Accumulate Unit Architectures for Deep Neural Network Accelerators[8]" In deep neural network accelerators, offers novel, low-complexity architectures for precisely scalable Multiply-Accumulate (MAC) units. Adding insights into effective hardware solutions for deep learning applications, this work is a significant contribution to the literature review in the subject of neural network accelerator design and optimization.

The work "Feed forward-Cut set-Free Pipelined Multiply and Accumulate Unit for Machine Learning Accelerators" by S. Ryu et al. presents an innovative pipelined Multiply-Accumulate (MAC) unit designed specifically for machine learning accelerators in order to do away with the necessity for feed forward cut sets [9]. This work provides important insights into optimizing hardware designs and represents a noteworthy step forward in the field of hardware acceleration for machine

learning. For academics and engineers looking to remain current on advancements in hardware acceleration for machine learning applications, it is an invaluable resource.

A new Time-Domain Multiply-Accumulate (MAC) unit is introduced in "Time-Domain Multiply-Accumulate Unit," written by P. S. Locatelli et al. [10]. Researchers and engineers interested in the latest developments in VLSI system design will find this work to be a useful resource as it adds to a wealth of literature on VLSI design and offers insight into a novel MAC unit design methodology. Z. Luo and colleagues' article [11]. The study "Accelerating Pipelined Integer and Floating-Point Accumulations in Configurable Hardware with Delayed Addition Techniques" looks into ways to use delay addition methods to speed up floating-point and integer based accumulation operations in configurable devices.

The design and real-world implementation of a Multiply-Accumulate (MAC) unit specialised for vector and Single Instruction, Multiple Data (SIMD) processing[12] is outlined in the article, "Architecture and Implementation of a Vector/SIMD Multiply-Accumulate Unit," authored by A. Danysh and associates. For researchers and other professionals seeking hardware design solutions to improve the effectiveness of vector and SIMD computations, it is a vital resource. An effective method for multiple-precision floating-point fused multiply-add operations with mixed-precision support is presented in the work "Efficient Multiple-Precision Floating-Point Fused Multiply-Add with Mixed-Precision Support," written by H. Zhang et al. [13].

III. PROPOSED METHOD

The process to generate a Multiply and Accumulate (MAC) unit involves several essential elements and phases. A multiplier, which is essential to the multiplication of two input values, is at the heart of a MAC unit. An accumulator is involved in this multiplication process; it is responsible for storing and continually accumulating the results. Registers are used to assist the management of input data and coefficients; data registers hold the input values, while coefficient registers hold the coefficients of the multiplier[14]. We will investigate two novel ways to execute out Multiply-Accumulate (MAC) operations in the section that follows. The LUT-Based MAC[15] and the LUT-Less MAC structures are the names of these techniques. We will provide insightful information by illuminating its distinct qualities[16] and capabilities through an extensive discussion.

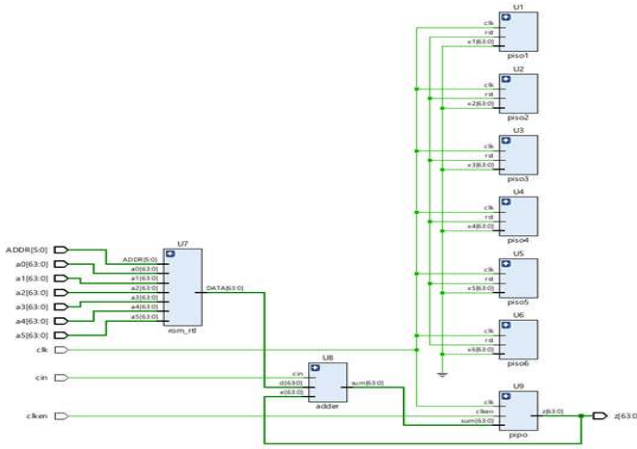


Fig. 2. RTL Schematic of LUT based 64 bit MAC

Fig. 2 is a Lookup Table [17] (LUT) Based Multiply-Accumulate (MAC) structure is a fundamental component in digital signal processing, particularly in applications like digital filtering, image processing, and various mathematical operations. This structure is essential for performing high-speed, efficient, and accurate computations in digital systems.

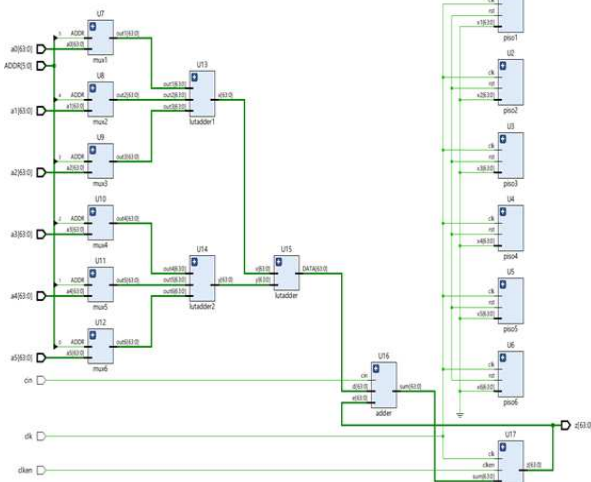


Fig. 3. RTL Schematic of LUT-LESS based 64 bit MAC

An inventive method for digital signal processing that aims to lessen or do away with the need for conventional Lookup Tables when executing Multiply-Accumulate (MAC) operations is the LUT-less (Lookup Table-less) based MAC structure is shown in Fig. 3. The primary goals of this approach are resource optimisation and increased system efficiency in digital systems[18].

In this work, we performed a thorough investigation of the performance of LUT-Based and LUT-Less based Multiply-Accumulate (MAC) structures, taking into account their power consumption and area utilization across various bit-width

configurations. The outcomes [19] we attained offer insightful information on the trade-offs between these two strategies[20][21].

IV. RESULTS AND DISCUSSION

The Simulation outcomes for a Multiply-Accumulate (MAC) process based on a LUT, and LIUT-Less, where the inputs are labeled a0, a1, a2, a3, a4, and a5, and the output is labeled Z as shown in Fig. 4 & Fig. 5.

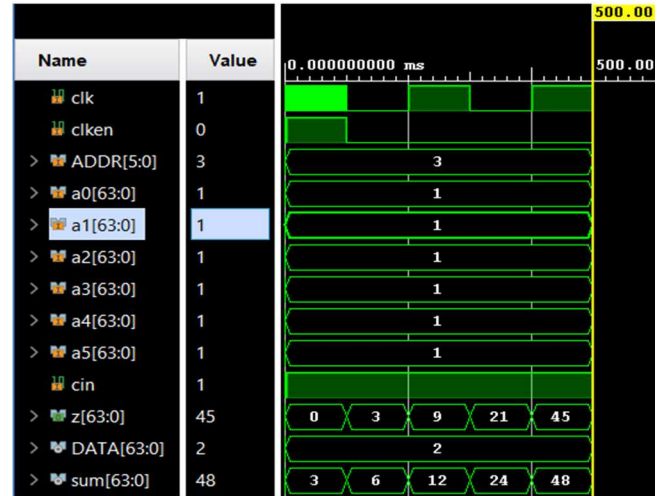


Fig. 4. Simulation Results of of LUT based 64 bit MAC

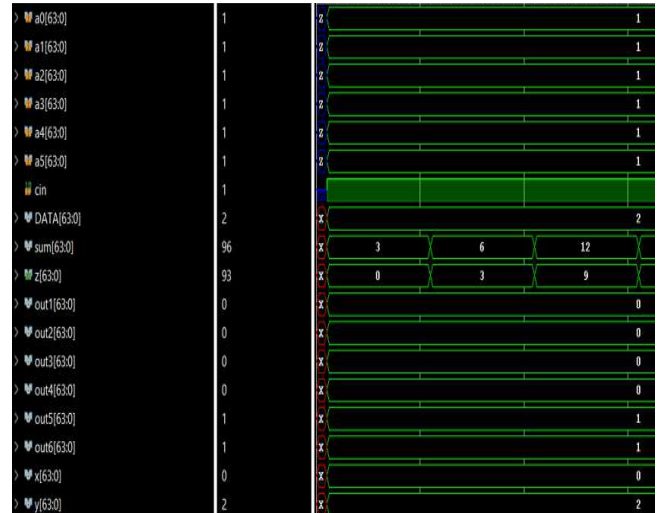


Fig. 5. Simulation Results of of LUT-Less based 64 bit MAC

. A Look-Up Table (LUT) is usually used in a LUT-based MAC operation to carry out multiplication and accumulation operations based on certain input values or addresses. In order to comprehend the consequences and significance of our research, we will look closely at the results and have a thorough conversation about them in this part. Using the Distributed Arithmetic approach, we have successfully built scalable MAC structures at different bit-widths: 4-bit, 8-bit, 16-bit, 32-bit, and 64-bit MAC structures.

We thoroughly investigated the performance metrics in a variety of bit-width as shown in TABLE 1. With regard to 4-bit structures, 737 units of area were utilised, and 4.833 watts of power were consumed. As one steps up to the 8-bit, there was a minor rise in area utilisation, coming to 810 units, and a matching logic power of 5.946 watts.

We found that as we moved into the 16-bit, there was a noticeable increase in both the amount of area needed—1390 units—and the amount of power needed—1120 watts. The tendency was further highlighted by the switch to 32- and 64-bit systems, which resulted in a significant increase in area and power measurements. In particular, the space usage increased to 2563 units for 32-bit structures, and the logic power consumed was 21.763 watts. Ultimately, the 64-bit required even more resources, requiring 3704 units of area occupation and 32.75 watts of electricity and as shown in Fig. 6 and Fig. 7. These findings underline the complex link between bit-width, area utilisation, and power consumption in our study and provide a crucial benchmark for our investigation.

TABLE 1: COMPARISON OF LUT VS LUT-LESS BASED MAC FOR VARIOUS BIT WIDTHS

Design	LUT BASED MAC		LUT-Less BASED MAC	
	AREA(LUT's)	LOGIC POWER(W)	AREA(LUT's)	LOGIC POWER(W)
4 bit	737	4.833	203	0.601
8 bit	810	5.946	214	0.867
16 bit	3.51390	11.20	262	1.557
32 bit	2563	21.763	373	2.527
64 bit	3704	32.75	512	3.590

When it came to 4-bit configurations, the LUT-Less MAC architecture demonstrated exceptional performance, requiring only 203 area units and 0.6 watts of logic power at most. It was able to keep its efficiency when we switched to the 8-bit LUT-Less architecture, requiring 0.867 watts of logic power instead of 214 area units. The LUT-Less technique held its ground and showed competitive performance numbers even with 16-bit MAC setups. In this instance, it used 262 area units and required 1.557 watts of logic power. As we went up to 32-bit systems, this trend towards efficiency continued, with the LUT-Less MAC structure using 373 area units and consuming 2.527 watts of logic power. The present analysis highlights notable benefits in terms of area and power efficiency when comparing a LUT-Based MAC design with a LUT-Less MAC design as indicated in TABLE 1. Compared to the LUT-Based MAC, which requires 3704 LUTs, the LUT-Less MAC design requires just 512 LUTs, an incredible 86.15% reduction in LUT area. For integrated circuit design, this significant hardware resource reduction is very helpful because it preserves valuable silicon area and may also lower manufacturing costs.

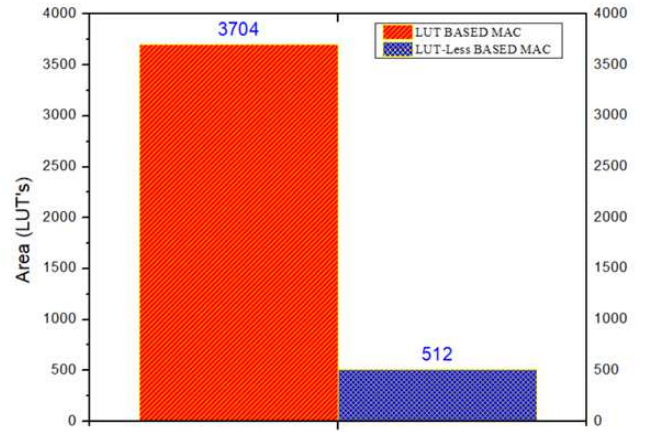


Fig. 6. Area Analysis of 64 bit LUT Vs LUT-Less based MAC structures

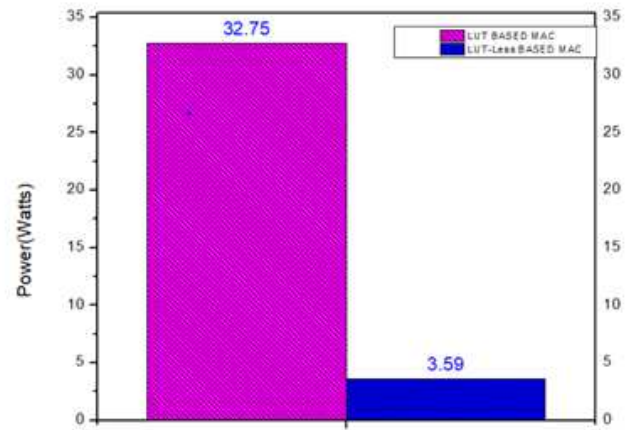


Fig. 7. Power Analysis of 64 bit LUT Vs LUT-Less based MAC structures

Furthermore, in terms of logic power consumption, the LUT-Less MAC design performs quite well. It uses only 3.590 watts to function, which is around 88.89% less than the 32.75 watts that the LUT-Based MAC needs. In many applications, lower logic power consumption is very beneficial, especially for battery-powered devices and systems with tight power budgets. Because of its effectiveness in this area, the LUT-Less MAC is a great option for developers and designers who want to maximise both area and power efficiency in their hardware implementations[22]. It is especially well-suited for portable and energy-conscious applications.

V. CONCLUSION

According to our findings, applications where low power consumption and resource constraints are critical are especially well-suited for LUT-Less MAC designs. These structures offer a strong substitute for high-performance MAC operations in addition to being exceptionally efficient and versatile. LUT-Based MAC architectures, on the other hand, have higher resource and power requirements despite being thorough and

accurate. The unique needs and limitations of the intended application should be taken into account while selecting between the two methods.

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